This method is considered as the effective one in regions where the availability of water is less.



Figure 21.10 Drip Irrigation System

DO YOU KNOW?

The global population is expected to be 9 billion by the year 2050. But, agriculture activities alone utilize 70% of the available fresh water resources. So, efficient and sustainable water use is needed for our own generation and future generations. Drip irrigation is a better solution for economical use of water.

21.2.5 Weeding

In an agriculture field, many other undesirable plants may grow naturally along with the main crop. These undesirable plants are called **weeds**. The removal of weeds is called weeding. Weeding is an important process because weeds compete with the crop plants for the nutrients, sunlight, water, space and other resources. It results in the under nourishment of crops and leads to low yield. It is mandatory to remove seeds from the field to achieve the expected yield. Farmers adopt many ways to remove weeds and control their growth. Some of them are explained below.

Mechanical methods

This is the most common method in which weeds are destroyed physically. Hand pulling or weeding with the help of weeding hoe is the oldest and most efficient method for controlling weeds.

Tillage methods

It is one of the practical methods of destroying weeds of all categories. Weeds are buried in the soil and also exposed to sun heat by deep ploughing.

Crop rotation

In this method, proper rotation of crops is followed for controlling crop associated and parasitic weeds.

Summer tillage

Deep ploughing after harvest of rabi crop and exposing underground parts of weeds to strong sunlight during summer months is useful for destroying many annual and perennial weeds.

Biological weed control

In this method, bio agents like insects and pathogens are used to control weeds. The objectives of biological control are not eradication, but reduction and regulation of the weed population.

Chemical methods

Chemical methods are very effective in certain cases and have great scope in weed control. The chemicals used for killing the weeds or inhibiting their growth are called **herbicides.** These chemicals are mixed with water and sprayed over the crops.



Figure 21.11 Weeding methods

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